

Anshuman Pandey

MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

CSE (AI/ML) student at VIT Bhopal with hands-on experience across the full ML lifecycle — from model development to production deployment and monitoring. Built end-to-end MLOps pipelines covering feature engineering, experiment tracking, model serving via FastAPI, containerization with Docker, and automated CI/CD with drift monitoring. Designed a statistically rigorous A/B testing platform for a recommender system, and engineered a chess engine using MCTS and reinforcement learning. Proficient in Python, PyTorch, FastAPI, Docker, and MLOps tooling (MLflow, DVC, GitHub Actions, Evidently AI). Actively seeking internship or entry-level Machine Learning Engineer roles.

EXPERIENCE

ServiceNow Virtual Intern | May 2026

- Completed a virtual internship focused on ServiceNow platform administration and automation, gaining practical exposure to Flow Designer, Automated Test Framework (ATF), Reporting, and Agentic AI concepts.

PROJECTS

MLOps Demand Forecasting Pipeline | Python, LightGBM, FastAPI, DVC, MLflow, Evidently AI, Docker | 2026

- Built an end-to-end MLOps pipeline for spatio-temporal traffic demand forecasting, covering data versioning, feature engineering, model training, REST API serving, and automated drift monitoring.
- Engineered temporal and cyclical features (geohash encoding, rush-hour flags, sine/cosine hour encoding) and trained a LightGBM regression model, logging RMSE, MAE, and R² via MLflow tracked on DagsHub.
- Deployed the model as a FastAPI/Uvicorn REST endpoint for real-time demand prediction, containerized with Docker.
- Implemented Evidently AI drift monitoring and a GitHub Actions CI/CD pipeline running automated unit tests on every push to main, and versioned datasets and artifacts with DVC on a Google Drive remote.

<https://github.com/AnshumanJ28/mlops-demand-forecasting>

YOLOv8 + ByteTrack Multi-Object Tracking | Python, Ultralytics YOLOv8, ByteTrack, OpenCV, Supervision, Gradio, MLflow | 2026

- Built a real-time object detection and multi-object tracking pipeline, combining YOLOv8 detection with ByteTrack to assign persistent tracking IDs across video frames.
- Generated annotated output video with bounding boxes, class labels, confidence scores, and track IDs, applicable to traffic monitoring, surveillance, and retail analytics use cases.
- Exposed the pipeline through a Gradio interface for interactive testing and tracked experiments with MLflow.

<https://github.com/AnshumanJ28/YoLo8>

AlphaZ0 — Neural Chess Engine | Python, PyTorch, MCTS, A3C, Pygame | 2026

- Designed a ResNet dual-head neural network (10 residual blocks, 128 channels) with separate policy and value heads, modeled after DeepMind's AlphaZero architecture, and built a fully playable engine with an 18-plane board encoder.
- Implemented MCTS with UCB selection and Dirichlet noise at root, combined with a parallel A3C training loop where worker threads compute GAE-advantage gradients asynchronously — engineered for concurrent, scalable self-play training.

<https://github.com/AnshumanJ28/AlphaZ0>

TECHNICAL SKILLS

MLOps & Deployment: FastAPI, Docker, DVC, MLflow, GitHub Actions (CI/CD), Evidently AI, LightGBM

ML / AI: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, CNN, LSTM, YOLO8

Languages: Python, C, C++, Java, SQL (MySQL), HTML, CSS

Tools & Platforms: AWS, Git, Docker, Jupyter Notebook, Google Colab, Kaggle Notebook, N8N

Concepts: Data Structures & Algorithms, Statistical Testing, System Design fundamentals

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — Vellore Institute of Technology, Bhopal

2023 – Present | CGPA: 8.52/10 • Class XII: 89.5% • Class X: 89.4%

CERTIFICATIONS

C & C++ Programming (Udemy, HackerRank)|**Python & Java** (HackerRank)|**Cloud Computing** (NPTEL)|**Advanced Excel** (Udemy) |**AI & Agentic AI** (TalentSpring)|**ML** (Coursera)|**AWS Trainer & AWS Academy** (AWS)|**IoT** (NPTEL)|**ServiceNow**

ACHIEVEMENTS

- 2nd Runner-up**, IEEE-NSUT Ideathon — awarded for ML-based project among 80+ competing teams.
- Top 10 finish**, BITS Pilani Hackathon — among top teams nationally in a competitive problem-solving event.